



The Papyrus Handbook

A guide to Markdown books with Papyrus

milon/papyrus

The Papyrus Handbook

Markdown to PDF, EPUB, HTML sites, and KDP

Nuruzzaman Milon

Welcome

Thanks for opening **The Papyrus Handbook**. Use the sidebar to browse chapters, or continue from Introduction. The home page has the project banner, a **Downloads** link for PDF previews from GitHub, and links to GitHub and Packagist.

This guide is the companion for [milon/papyrus](#) — a PHP CLI that turns Markdown into PDF, EPUB, HTML, a multi-page site, and Amazon KDP exports. You are reading those same chapters as a Papyrus site.

How to use this guide

1. **Introduction** — what Papyrus is and the project layout
2. **Install** — Composer setup and `papyrus init`
3. **Configuration** — every `papyrus.php` option
4. **Writing** — Markdown, asides, Mermaid, linting
5. **Themes** — PDF / HTML themes and fonts
6. **build** — all-in-one export, watch, caches
7. **build:pdf / epub / html / site / sample** — one chapter each
8. **kdp** and **kdp:*** — Kindle, print, covers, metadata
9. **Migration and CI** — ibis-next migration and GitHub Actions
10. **Command reference** — full CLI tables
11. **Downloads** — light and dark PDF previews from GitHub

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Introduction

Papyrus is a Composer package and CLI for authors who write books in Markdown and need more than one delivery format from a single tree of chapters.

One project layout yields:

- **PDF** — print interiors and shareable digital books
- **EPUB** — stores, e-readers, and Kindle upload
- **HTML** — a single file with light / dark mode
- **Site** — a multi-page static website (this handbook)
- **KDP helpers** — Kindle EPUB, print PDF, covers, metadata JSON

This handbook is itself a Papyrus book. The Markdown source lives in [examples/the-papyrus-handbook/](#) on GitHub. Prebuilt PDF, HTML, and site outputs are published from that example so you can read without installing anything.

Who it is for

- Authors shipping a technical or long-form book from Markdown
- Teams migrating off [ibis-next](#)
- Anyone who wants PDF and a hosted docs site from the same chapters

Project shape

Every book is a directory with:

Path	Role
<code>papyrus.php</code>	Title, trim, themes, Mermaid, samples, KDP
<code>content/</code>	Markdown chapters (natural filename order)
<code>assets/</code>	PDF / HTML themes, covers, fonts, CSS
<code>export/</code>	Build outputs (usually gitignored)

Next steps

Install Papyrus (next chapter), run `papyrus doctor`, then `papyrus build` or `papyrus build:site`. Later chapters document every build and KDP command and the full `papyrus.php` option set.

Source and releases: github.com/milon/papyrus.

Install and project layout

Requirements

Required

- PHP 8.2+ with extensions `dom`, `gd`, `mbstring`, `zip`, and `zlib`
- Composer

Optional

Tool	Used by	Notes
<code>mmdc</code> (<code>@mermaid-js/mermaid-cli</code> or Homebrew <code>mermaid-cli</code>)	Mermaid diagrams	Needs Chrome/Chromium for Puppeteer
Chrome or Chromium	Mermaid CLI	Set <code>PUPPETEER_EXECUTABLE_PATH</code> if needed
<code>epubcheck</code>	<code>kdp:ebook</code>	Extra validation; skipped with a warning when absent

macOS (Homebrew)

```
brew install php composer
brew install mermaid-cli
brew install --cask google-chrome
brew install epubcheck
```

```
export PUPPETEER_EXECUTABLE_PATH="/Applications/Google  
Chrome.app/Contents/MacOS/Google Chrome"
```

npm / Linux

```
npm install -g @mermaid-js/mermaid-cli  
# Debian/Ubuntu example:  
sudo apt-get install -y chromium-browser  
export PUPPETEER_EXECUTABLE_PATH="$(command -v chromium-browser  
|| command -v google-chrome || command -v chromium)"
```

Install epubcheck and put it on **PATH**, or use **brew install epubcheck** on macOS.

Check the toolchain

```
php -m | grep -E 'dom|gd|mbstring|zip|zlib'  
mmdc --version  
epubcheck --version  
papyrus doctor
```


Install per project

In your book repository:

```
composer require milon/papyrus
```

```
vendor/bin/papyrus --version  
vendor/bin/papyrus init  
vendor/bin/papyrus doctor
```

Wire Composer scripts (example):

```
{  
  "scripts": {  
    "build": "papyrus build",  
    "build:pdf": "papyrus build:pdf --theme light,dark",  
    "build:epub": "papyrus build:epub",  
    "build:html": "papyrus build:html",  
    "build:site": "papyrus build:site",  
    "build:sample": "papyrus build:sample",  
    "build:kdp": "papyrus kdp",  
    "doctor": "papyrus doctor"  
  }  
}
```

Package page: packagist.org/packages/milon/papyrus.

Install globally

```
composer global require milon/papyrus  
export PATH="$(composer global config bin-dir --absolute):$PATH"  
papyrus list
```

Scaffold a book

```
papyrus init  
papyrus init -d my-book
```

`init` creates `papyrus.php`, `content/`, and `assets/`. Use `--force` / `-f` to overwrite.

Layout

Path	Role
<code>papyrus.php</code>	Book settings
<code>content/</code>	Markdown chapters
<code>assets/</code>	Themes, CSS, covers, fonts
<code>export/</code>	Built artifacts
<code>.papyrus/</code>	Incremental caches

Always run from the book root, or pass `-d / --dir`. Override where artifacts are written with `-e / --export` (default: `<book>/export`):

```
papyrus doctor -d /path/to/book
papyrus build --dir /path/to/book
papyrus build:site -d /path/to/book -e /path/to/out
```

Browse the Papyrus source on [GitHub](#) if you want to follow along with this handbook's own project under `examples/the-papyrus-handbook/`.

Configuration

All settings live in `papyrus.php`, which returns a PHP array.

Identity and paths

```
return [
    'title' => 'My Book',
    'subtitle' => 'A short subtitle',
    'author' => 'Author Name',
    'language' => 'en',
    'themes' => ['light', 'dark'],

    'content_dir' => 'content',
    'assets_dir' => 'assets',
    'export_dir' => 'export',
];
```

Key	Default	Notes
<code>title</code>	<code>Untitled</code>	Drives the export filename slug
<code>subtitle</code>	<code>''</code>	
<code>author</code>	<code>''</code>	
<code>language</code>	<code>en</code>	EPUB / KDP metadata fallback
<code>themes</code>	<code>['light', 'dark']</code>	Each needs <code>assets/theme-{name}.html</code>
<code>content_dir</code>	<code>content</code>	Relative to the book root

Key	Default	Notes
<code>assets_dir</code>	<code>assets</code>	
<code>export_dir</code>	<code>export</code>	Overridden by <code>-e</code> / <code>--export</code>

The slug is the title lowercased with non-alphanumeric runs turned into `-`. This handbook is *The Papyrus Handbook*, so outputs look like `the-papyrus-handbook-light.pdf`.

Page size and margins

```
'document' => [  
  'size' => 'crown-quarto',  
  'margin_left' => 27,  
  'margin_right' => 27,  
  'margin_top' => 14,  
  'margin_bottom' => 14,  
],
```

List presets with **papyrus sizes**. Custom trim:

```
'document' => [  
  'format' => [152.4, 228.6], // width_mm, height_mm  
],
```


Table of contents

```
'toc' => [  
  'h1' => 0,  
  'h2' => 0,  
  'h3' => 1,  
],
```

Values follow mPDF **h2toc** conventions used by Papyrus (**0** include, **1** exclude).

Cover images

```
'cover' => [  
  'image' => 'cover.png',  
  'light' => 'cover-light.png',  
  'dark' => 'cover-dark.png',  
  // 'width' => 188.976, // mm; defaults to page width  
  // 'height' => 246.126, // mm; defaults to page height  
],
```

Paths are relative to **assets/**. Missing per-theme covers fall back to **cover.image**. KDP print PDFs omit the cover page entirely.

Running header

```
'header' => [  
  'style' => 'font-style: italic; text-align: right; border-  
bottom: solid 1px #808080;',  
],
```

Pretoc chapters suppress folios differently from body chapters.

Fonts

Declare faces under `assets/fonts/` and optional **script routing** (PDF only). First matching **script** rule wins.

```
'fonts' => [  
  'default' => 'librelibertine', // optional; else first face,  
  // else mPDF default  
  'faces' => [  
    [  
      'name' => 'librelibertine',  
      'regular' => 'LinLibertine_R.ttf',  
      'bold' => 'LinLibertine_RB.ttf',  
      'italic' => 'LinLibertine_RI.ttf',  
      'bold_italic' => 'LinLibertine_RBI.ttf',  
    ],  
    [  
      'name' => 'notosansbengali',  
      'regular' => 'NotoSansBengali-Regular.ttf',  
      'otl' => true, // OpenType layout – needed for  
      // complex scripts  
    ],  
  ],  
  'script' => [  
    // When mPDF tags a run as Bengali, use this face instead  
    // of the default  
    ['match' => ['bn', 'ben', 'bengali'], 'face' =>  
    'notosansbengali'],  
  ],  
],
```

Faces

Each face needs a **name** and a **regular** file that exists under **assets/fonts/**. Optional keys: **bold**, **italic**, **bold_italic**, and **otl** (sets mPDF **useOTL** when true — common for code fonts and Indic / Arabic scripts).

Papyrus merges these into mPDF's **fontdata** when building PDF (**FontRegistry** → **MpdfFactory**).

Script rules

fonts.script is a list of { **match**, **face** } rules used only for **PDF**. They do not change EPUB or HTML typography.

When any applicable rule exists, Papyrus turns on mPDF's **autoScriptToLang** and **autoLangToFont**, and installs a custom **languageToFont** resolver (**ScriptLanguageToFont**). mPDF then:

1. Detects script runs in the HTML (e.g. Bengali glyphs).
2. Tags them with a language / script code (e.g. **bn**, or ***-beng**).
3. Asks Papyrus which face to use; the first rule whose **match** list hits that code wins.

match aliases are lowercased. Built-in aliases include:

Aliases you can list	Resolves as
bn , ben , beng , bengali	Bengali

Aliases you can list	Resolves as
<code>ar</code> , <code>arab</code> , <code>arabic</code>	Arabic
<code>hi</code> , <code>hin</code> , <code>deva</code> , <code>devanagari</code> , <code>hindi</code>	Devanagari
any 4-letter ISO script code	that script as-is

The `face` string must be a name from `fonts.faces` and that face must actually load (regular file present). Rules pointing at a missing face are skipped (`FontRegistry::applicableScriptRules()`). Unmatched languages fall through to mPDF's built-in `LanguageToFont`.

Example: body text in Latin uses `librelibertine`; a paragraph containing বন্ধন is auto-tagged Bengali and rendered with `notosansbengali` if that face is registered.

Notice: Without both a registered face and a matching `script` rule, non-Latin runs often show as tofu (missing glyphs) in the default font.

Site home (**build:site**)

```
'site' => [  
  'banner' => 'banner.jpg',  
  'repository' => 'https://github.com/you/your-book',  
  'lead' => 'A one-line pitch for the home page.',  
],
```

Key	Default	Notes
banner	auto banner.jpg / banner.png if present	Under assets/
repository	unset	Repo URL; GitHub hosts also surface Packagist / Issues links
lead	unset	Home page pitch

Mermaid

```
'mermaid' => [
  'enabled' => true,
  'format' => 'svg',           // svg | png
  'theme' => 'auto',          // auto = book colours; or
                              default|dark|forest|neutral
  'max_width_mm' => 130,
  // 'command' => '/usr/local/bin/mmdc', // optional override
],
```

Key	Default	Notes
enabled	false	Requires mmdc on PATH (or command)
format	svg	Anything other than png becomes svg
theme	auto	Book-matched palettes for PDF; HTML/site embed light + dark
max_width_mm	130	PDF figure width cap
command	auto-resolve mmdc	Explicit CLI path

Sample PDF

```
'sample' => [  
  'ranges' => [  
    ['from' => 1, 'to' => 3],  
  ],  
,  
  'sample_notice' => 'This is a sample from My Book.',
```

Ranges are 1-based inclusive pages of the finished theme PDF. Notice text also accepts `sample.notice` / `sample.text` as fallbacks. See the `build:sample` chapter.

KDP

```
'kdp' => [  
  'ebook' => [  
    'enabled' => true,  
    'cover' => 'cover-ebook.jpg',  
  ],  
  'print' => [  
    'enabled' => true,  
    'bleed_mm' => 3,  
    'margin_preset' => 'recommended', // or minimum  
    'paper' => 'white',  
  ],  
  'metadata' => [  
    'description' => 'Bookstore blurb...',  
    'keywords' => ['laravel', 'filament'],  
    'language' => 'en',  
  ],  
],
```

Full option tables live in the **kdp** and **kdp:*** chapters. **papyrus build** only runs KDP when ebook or print is enabled; **papyrus kdp** errors if neither is.

Break level

```
'break_level' => 2,
```

Automatic page breaks before headings (**1** = H1, **2** = H1 and H2). Explicit **[break]** markers always insert a break.

CommonMark hook

```
'configure_commonmark' => function  
(\League\CommonMark\Environment\Environment $environment): void {  
    // register custom extensions  
},
```

Writing content

Chapters are Markdown under `content/`, loaded in natural filename order (`00-...`, `01-...`, `10-...`). This handbook's Welcome page is a pretoc chapter; everything after Introduction is body content for the PDF table of contents.

Front matter

```
---  
title: My chapter  
pretoc: true  
---  
  
Body starts here.
```

Key	Meaning
<code>title</code>	Chapter title for EPUB / internal use
<code>pretoc</code>	<code>true</code> places the chapter before the PDF TOC

Markdown features

CommonMark + GFM, fenced code highlighting, and book extras.

Emphasis and code

```
Hello, world.

`inline code`

```php
$user = User::factory()->create();
```
```

Callouts

```
> {notice} Something helpful.
> {warning} Something risky.
```

```
::: note
A note aside.
:::

::: warning
A warning aside.
:::
```

Page breaks

```
[break]
```

Mermaid

Write a **mermaid** fence in your chapter. Papyrus runs **mmdc** at build time and replaces the fence with an SVG (or PNG) figure in PDF, EPUB, HTML, and site exports.

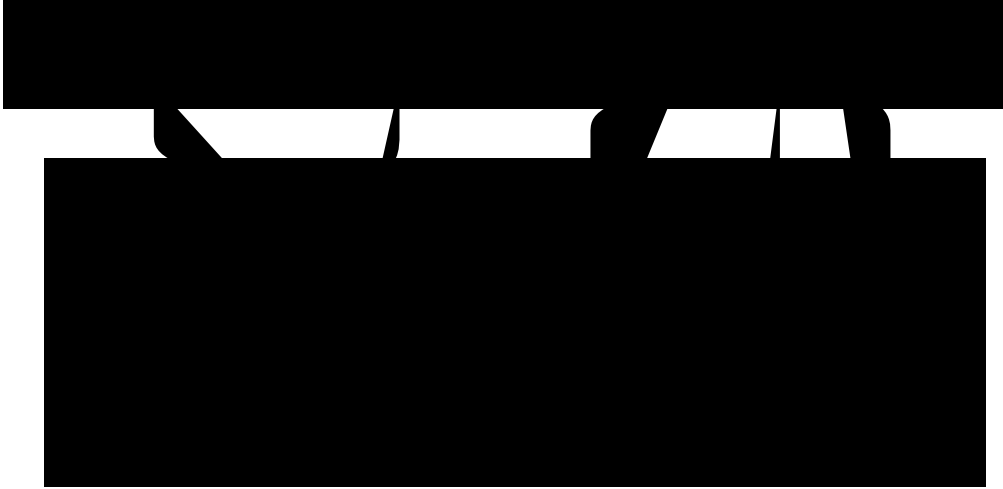
Source:


```
```mermaid
flowchart TB
 subgraph Authoring
 MD[Markdown chapters]
 CFG[papyrus.php]
 ASSETS[Themes and fonts]
 end

 subgraph Papyrus
 CONVERT[BookConverter]
 MERMAID[MermaidRenderer]
 PDF[build:pdf]
 EPUB[build:epub]
 HTML[build:html]
 SITE[build:site]
 end

 MD --> CONVERT
 CFG --> CONVERT
 ASSETS --> PDF
 ASSETS --> HTML
 ASSETS --> SITE
 CONVERT --> MERMAID
 MERMAID --> PDF
 MERMAID --> EPUB
 MERMAID --> HTML
 MERMAID --> SITE
```
```

Rendered output:



Enable in `papyrus.php` (theme defaults to `auto` — book colours; HTML and site embeds both light and dark variants):

```
'mermaid' => [  
    'enabled' => true,  
],
```

Optional knobs: `format` (`svg` / `png`), `theme` (`auto` or a Mermaid stock theme like `default` / `dark` / `forest`), `max_width_mm`.

Requires `@mermaid-js/mermaid-cli` (`mmdc`) on `PATH`. Diagrams cache under `.papyrus/cache/mermaid`.

PHP fence linting

```
papyrus lint
papyrus lint --fix
papyrus lint --max-width=66
```

| Option | Short | Default | Meaning |
|-------------------------------|----------------------|-----------------|--|
| <code>--fix</code> | <code>-f</code> | off | Apply auto-fixes for open tags and comment runs |
| <code>--max-width</code> | — | <code>66</code> | Maximum line width for PHP fences |
| <code>--dir / --export</code> | <code>-d / -e</code> | book defaults | Shared book options (<code>--export</code> unused for lint) |

Themes and assets

Everything visual lives under `assets/`. New projects from `papyrus init` ship a Filament Playbook–inspired theme: Linux Libertine for body and headings, 0xProto for code, github-gist highlight colors, and notice / tip / caution asides, plus matching font files under `assets/fonts/`.

PDF themes

For each name in `themes`, provide `assets/theme-{name}.html`. Split the theme on the TOC marker:

```
<!-- PAPYRUS:TOC -->
```

Papyrus writes pretoc chapters, then the TOC, then body chapters around that marker. A legacy `<!-- IBIS:TOC -->` marker is still recognized after `migrate-ibis`.

Typical theme responsibilities: `@page` / typography, code blocks, callout colors, title page, and running header styles (often paired with `header.style` in config).

HTML theme

`assets/theme-html.html` is a full HTML document with placeholders `{{title}}`, `{{subtitle}}`, `{{author}}`, and `{{body}}`. The default template mirrors the light PDF theme (same typefaces, colors, code blocks, and callouts), with a fixed moon/sun icon toggle that switches to the dark PDF palette. The reading column follows Tailwind screen widths (640 → 1536px). `@font-face` rules in the theme use `../assets/fonts/`; at build time Papyrus rewrites those URLs relative to the HTML output directory (so `-e docs` still finds the book's fonts).

```
papyrus build:html
papyrus build:site
```

`build:site` writes `export/<slug>-site/` with one HTML page per chapter, a Home index, shared CSS/JS (fonts copied into the site), a collapsible chapter sidebar, and the same light/dark toggle as the single-file HTML export. That folder is what this handbook deploys to GitHub Pages.

EPUB styles

`assets/style.css` (and optional `highlight.codeblock.min.css`) ship inside the EPUB. Prefer simple `pre/code` rules for e-ink readers.

Covers and fonts

Place cover images and font files under `assets/` (fonts usually in `assets/fonts/`). Reference them from `cover` and `fonts.faces` in `papyrus.php`. This handbook's cover is `assets/cover.jpg`; the wide banner used on the Welcome page is published from the Papyrus repo at [assets/papyrus-banner.jpg](#).

build

papyrus build is the all-in-one export command. From the book root (or with **-d**), it runs every configured PDF theme, then EPUB, then single-file HTML, then every **enabled** KDP output.

```
papyrus build
papyrus build -d /path/to/book
papyrus build -d examples/the-papyrus-handbook -e docs
```

Options

| Option | Short | Default | Meaning |
|-----------------------|-----------------|---|--|
| <code>--dir</code> | <code>-d</code> | current directory | Book root (must contain <code>papyrus.php</code>) |
| <code>--export</code> | <code>-e</code> | <code>export_dir</code> from config (<code>export</code>) | Where artifacts are written |

Global Symfony options also apply (`-v` / `--verbose`, `-q`, `--no-ansi`, ...).
With `-v`, Papyrus prints third-party notices from mPDF and the EPUB zipper.

What it builds

| Step | Always? | Output |
|---|---|---------------------------|
| PDF
(each
themes
entry) | yes | export/<slug>-<theme>.pdf |
| EPUB | yes | export/<slug>.epub |
| HTML | yes | export/<slug>.html |
| KDP
ebook /
print /
covers /
metadata | only if
kdp.ebook.enabled
or
kdp.print.enabled | see KDP chapters |

Slug comes from **title** (lowercased, non-alphanumeric → -).

What it does *not* build

These need their own commands:

- `build:site` — multi-page static site
- `build:sample` — sample PDF from page ranges

`watch` also invokes `build` only (not site or sample).

Failures

Each step is independent. If one format fails, Papyrus continues with the others and exits non-zero if any step failed. When no KDP outputs are enabled, the KDP block is skipped silently (unlike `papyrus kdp`, which errors).

Validate first

```
papyrus doctor  
papyrus build
```

Watch mode

Rebuild on changes to `papyrus.php`, `content/`, and `assets/`:

```
papyrus watch  
papyrus watch --interval 3
```

| Option | Short | Default | Meaning |
|-------------------------------|----------------------|----------------------------|--|
| <code>--interval</code> | <code>-i</code> | <code>2</code> | Poll interval in seconds (minimum <code>1</code>) |
| <code>--dir / --export</code> | <code>-d / -e</code> | same as <code>build</code> | Passed through to each rebuild |

Caching

Chapter HTML caches under `.papyrus/cache/markdown`. Mermaid figures use `.papyrus/cache/mermaid`. Delete `.papyrus/` for a cold rebuild.

Related chapters

The next chapters cover each `build:*` command and every KDP command in turn.

build:pdf

Build one PDF per theme (or a subset). Each theme needs `assets/theme-{name}.html` and an entry in `themes`.

```
papyrus build:pdf
papyrus build:pdf --theme light
papyrus build:pdf --theme light,dark --parallel
papyrus build:pdf -d /path/to/book -e /path/to/out
```

Options

| Option | Short | Default | Meaning |
|-------------------------|-----------------|-----------------------|-----------------------------|
| <code>--dir</code> | <code>-d</code> | current directory | Book root |
| <code>--export</code> | <code>-e</code> | <code>export/</code> | Output directory |
| <code>--theme</code> | <code>-t</code> | all configured themes | Comma-separated theme names |
| <code>--parallel</code> | <code>-p</code> | off | One process per theme |

Output

```
export/<slug>-<theme>.pdf
```

Example for this handbook: **the-papyrus-handbook-light.pdf**.

Config that affects PDF

| Key | Role |
|--|---|
| <code>themes</code> | Which theme names to build by default |
| <code>document.size / document.format</code> | Trim size |
| <code>document.margin_*</code> | Margins in millimetres |
| <code>cover</code> | Cover image(s) under <code>assets/</code> |
| <code>header.style</code> | Running header CSS |
| <code>toc</code> | Which heading levels enter the TOC |
| <code>break_level</code> | Auto page breaks before headings |
| <code>fonts</code> | Faces and script routing |
| <code>mermaid</code> | Diagram rendering |

List trim presets with `papyrus sizes`. Custom trim:

```
'document' => [  
  'format' => [152.4, 228.6], // width_mm, height_mm  
],
```

Parallel builds

`--parallel` / `-p` spawns one PHP process per theme. Useful when `themes` has both `light` and `dark`. Sequential is the default (and what `papyrus build` uses).

Covers

```
'cover' => [  
  'image' => 'cover.png',          // fallback for every theme  
  'light' => 'cover-light.png',    // optional per-theme  
  'dark' => 'cover-dark.png',  
  // 'width' / 'height' - mm; default to page size  
],
```

Paths are relative to **assets/**. Missing per-theme covers fall back to **cover.image**.

build:epub

Build an EPUB3 package from the same Markdown chapters as PDF and HTML.

```
papyrus build:epub  
papyrus build:epub -d /path/to/book  
papyrus build:epub -e /path/to/out
```

Options

| Option | Short | Default | Meaning |
|-----------------------|-----------------|----------------------|------------------|
| <code>--dir</code> | <code>-d</code> | current directory | Book root |
| <code>--export</code> | <code>-e</code> | <code>export/</code> | Output directory |

Output

export/<slug>.epub

What goes in the package

- Chapter HTML derived from `content/`
- `assets/style.css` and optional `highlight.codeblock.min.css`
- Embedded images referenced from chapters
- Cover from `cover` (theme fallback → `cover.image`)
- Metadata from `title`, `subtitle`, `author`, and `language`

Config that affects EPUB

| Key | Role |
|---|--|
| <code>title</code> , <code>subtitle</code> ,
<code>author</code> | Package identity |
| <code>language</code> | EPUB language (default <code>en</code>) |
| <code>cover</code> | Cover image under <code>assets/</code> |
| <code>mermaid</code> | Pre-rendered figures |
| <code>fonts</code> | Faces are PDF-oriented; EPUB relies on CSS |

Prefer simple `pre` / `code` rules in `assets/style.css` for e-ink readers.

Kindle upload

For Amazon KDP's Kindle pipeline, prefer `kdp:ebook` (validates and names the file for upload). `build:epub` is the general-purpose store / archive build.

build:html

Build a **single-file** HTML book with light / dark mode.

```
papyrus build:html  
papyrus build:html -d /path/to/book -e docs
```

Options

| Option | Short | Default | Meaning |
|-----------------------|-----------------|----------------------|------------------|
| <code>--dir</code> | <code>-d</code> | current directory | Book root |
| <code>--export</code> | <code>-e</code> | <code>export/</code> | Output directory |

Output

```
export/<slug>.html
```

Requires `assets/theme-html.html`. Placeholders in the theme:

| Placeholder | Source |
|------------------|-------------------|
| {{ \$title }} | title |
| {{ \$subtitle }} | subtitle |
| {{ \$author }} | author |
| {{ \$body }} | Rendered chapters |

The default stub theme mirrors the light PDF palette, with a moon/sun toggle that switches to the dark PDF colours. The reading column follows Tailwind screen widths (640 → 1536px).

`@font-face` URLs that point at `../assets/fonts/` are rewritten at build time so `-e docs` (or any export override) still finds the book's fonts.

Config that affects HTML

| Key | Role |
|---|--|
| <code>title</code> , <code>subtitle</code> ,
<code>author</code> | Document chrome |
| <code>mermaid</code> | Figures; <code>theme</code> $\Rightarrow$ <code>auto</code> embeds light + dark SVGs |
| <code>fonts</code> | Loaded via the HTML theme's <code>@font-face</code> rules |

Site vs HTML

| Command | Result |
|-------------------------|---|
| <code>build:html</code> | One self-contained <code>.html</code> file |
| <code>build:site</code> | Multi-page folder with sidebar, Home, shared CSS/JS |

Use `build:html` for emailing or archiving a single document; use `build:site` for hosting.

build:site

Build a multi-page static site: Home index, one HTML page per chapter, sidebar navigation, light / dark mode, and Prev / Next links.

```
papyrus build:site  
papyrus build:site -d examples/the-papyrus-handbook -e docs
```

papyrus build does **not** run this command — site output is opt-in.

Options

| Option | Short | Default | Meaning |
|-----------------------|-----------------|----------------------|--------------------------------------|
| <code>--dir</code> | <code>-d</code> | current directory | Book root |
| <code>--export</code> | <code>-e</code> | <code>export/</code> | Parent directory for the site folder |

Output

```
export/<slug>-site/  
  index.html  
  404.html  
  <chapter-slug>.html  
  .nojekyll  
  assets/site.css  
  assets/site.js  
  assets/fonts/...  
  assets/<banner>    # when configured
```

Point any static host (GitHub Pages, Netlify, S3, ...) at that folder. **.nojekyll** tells GitHub Pages not to run Jekyll. GitHub Pages and Netlify serve **404.html** for missing URLs.

Site config

Optional **site** block in **papyrus.php**:

```
'site' => [
    'banner' => 'banner.jpg',           // under assets/; auto-
    detects banner.jpg / banner.png
    'repository' => 'https://github.com/you/your-book',
    'lead' => 'A one-line pitch for the home page.',
],
```

| Key | Default | Meaning |
|-------------------|---|--|
| banner | banner.jpg then
banner.png if
present | Hero image on Home |
| repository | unset | Repo URL; GitHub hosts
also get Packagist / Issues
links when detected |
| lead | unset | Short pitch under the title
on Home |

Hosting this handbook

In the Papyrus repository, pushes to **main** rebuild the handbook site and publish it to **gh-pages** via **.github/workflows/pages.yml**. Locally:

```
composer build:handbook
# or just:
papyrus build:site -d examples/the-papyrus-handbook -e docs
```


Mermaid on the site

With `mermaid.theme => auto`, each diagram embeds light and dark SVG variants; CSS swaps them with `data-theme`. Diagrams stay within the reading column (same max-width ladder as the HTML theme).

build:sample

Carve page ranges from a finished theme PDF into a **sample** PDF — useful for marketing downloads or review copies.

```
papyrus build:sample  
papyrus build:sample --theme light  
papyrus build:sample -t light,dark
```

papyrus build does **not** run this command.

Options

| Option | Short | Default | Meaning |
|-----------------------|-----------------|-----------------------|-----------------------------|
| <code>--dir</code> | <code>-d</code> | current directory | Book root |
| <code>--export</code> | <code>-e</code> | <code>export/</code> | Output directory |
| <code>--theme</code> | <code>-t</code> | all configured themes | Comma-separated theme names |

Config

```
'sample' => [
  'ranges' => [
    ['from' => 1, 'to' => 4],
    ['from' => 10, 'to' => 12],
  ],
  // Optional notice text (also accepted as sample.notice /
  sample.text):
  // 'notice' => 'This is a sample...',
],

'sample_notice' => 'This is a sample from My Book.'
```

| Key | Meaning |
|--|---|
| <code>sample.ranges</code> | List of <code>{from, to}</code> 1-based inclusive page ranges against the finished PDF |
| <code>sample_notice</code> | Prefixed notice page (preferred) |
| <code>sample.notice</code> /
<code>sample.text</code> | Fallbacks if <code>sample_notice</code> is empty |

Invalid ranges are dropped. `build:sample` requires at least one valid range or it fails.

Output

```
export/sample-<slug>-<theme>.pdf
```

Papyrus builds the full theme PDF first, then extracts the configured pages. When a notice is set, it appears as the first page of the sample.

kdp

papyrus kdp builds every **enabled** Amazon KDP artifact in one shot: Kindle EPUB (if enabled), print interior PDF (if enabled), cover asset copies, and the metadata JSON sidecar.

```
papyrus kdp
papyrus kdp -d /path/to/book -e /path/to/out
```

Unlike **papyrus build**, this command **fails** when neither **kdp.ebook.enabled** nor **kdp.print.enabled** is true.

Options

| Option | Short | Default | Meaning |
|-----------------------|-----------------|----------------------|------------------|
| <code>--dir</code> | <code>-d</code> | current directory | Book root |
| <code>--export</code> | <code>-e</code> | <code>export/</code> | Output directory |

Print uses the **first** configured theme (same as `kdp:print` without `--theme`).

Enable flags

```
'kdp' => [
  'ebook' => [
    'enabled' => true,
    'cover' => 'cover-ebook.jpg',
  ],
  'print' => [
    'enabled' => true,
    'bleed_mm' => 3,
    'margin_preset' => 'recommended', // or minimum
    'paper' => 'white',                // metadata only (e.g.
cream)
  ],
  'metadata' => [
    'description' => 'Bookstore blurb...',
    'keywords' => ['papyrus', 'markdown'],
    'language' => 'en',
  ],
],
```

| Key | Default | Notes |
|----------------------------|--------------------|----------------------------------|
| <code>ebook.enabled</code> | <code>false</code> | Gates ebook EPUB |
| <code>ebook.cover</code> | unset | Asset under <code>assets/</code> |
| <code>print.enabled</code> | <code>false</code> | Gates print PDF |

| Key | Default | Notes |
|----------------------------------|----------------------|-------------------------------------|
| <code>print.bleed_mm</code> | 3 | Added to trim and each margin |
| <code>print.margin_preset</code> | recommended | recommended or minimum |
| <code>print.paper</code> | white | Recorded in metadata only |
| <code>metadata.*</code> | see metadata chapter | Sidecar + EPUB description fallback |

Typical artifacts

| Artifact | Filename |
|--------------------|---|
| Kindle EPUB | <code><slug>-kdp.epub</code> |
| Print PDF | <code><slug>-kdp-print.pdf</code> |
| Ebook cover copy | <code><slug>-kdp-ebook-cover.<ext></code> |
| Print cover copies | <code><slug>-kdp-print-cover-<theme>.<ext></code> |
| Metadata | <code><slug>-kdp-metadata.json</code> |

Confirm trim with `papyrus sizes` and Amazon's current KDP print specs before uploading. The next chapters document each `kdp:*` command.

kdp:ebook

Build a Kindle-oriented EPUB for Amazon KDP upload.

```
papyrus kdp:ebook  
papyrus kdp:ebook -d /path/to/book -e /path/to/out
```

Requires `kdp.ebook.enabled => true`.

Options

| Option | Short | Default | Meaning |
|-----------------------|-----------------|----------------------|------------------|
| <code>--dir</code> | <code>-d</code> | current directory | Book root |
| <code>--export</code> | <code>-e</code> | <code>export/</code> | Output directory |

Output

```
export/<slug>-kdp.epub
```

Papyrus runs an internal KDP-oriented check and, when available on **PATH**, **epubcheck** (install with **brew install epubcheck**, or from [w3c/epubcheck](https://w3c.github.io/epubcheck/)).

Cover

```
'kdp' => [  
  'ebook' => [  
    'enabled' => true,  
    'cover' => 'cover-ebook.jpg', // under assets/  
  ],  
],
```

If `kdp.ebook.cover` is unset, Papyrus falls back to the light theme cover (`cover.light` or `cover.image`).

Metadata

Description falls back to "{title} - {author}" when `kdp.metadata.description` is empty. Language uses `kdp.metadata.language` when set, otherwise project `language` (default `en`).

Versus build:epub

| | build:epub | kdp:ebook |
|---------------------|--------------------------|------------------------------------|
| Filename | <slug>.epub | <slug>-kdp.epub |
| Enable flag | always | kdp.ebook.enabled |
| Validation | packaging only | KDP checks + optional
epubcheck |
| Cover
preference | theme cover | kdp.ebook.cover first |

kdp:print

Build a **print interior** PDF sized for Amazon KDP: no cover page, KDP margin presets, and bleed applied around the project trim.

```
papyrus kdp:print  
papyrus kdp:print --theme light  
papyrus kdp:print -t dark -e /path/to/out
```

Requires `kdp.print.enabled => true`.

Options

| Option | Short | Default | Meaning |
|-----------------------|-----------------|------------------------|-------------------------------|
| <code>--dir</code> | <code>-d</code> | current directory | Book root |
| <code>--export</code> | <code>-e</code> | <code>export/</code> | Output directory |
| <code>--theme</code> | <code>-t</code> | first configured theme | PDF theme for interior layout |

Output

```
export/<slug>-kdp-print.pdf
```

Print config

```
'kdp' => [
  'print' => [
    'enabled' => true,
    'bleed_mm' => 3,
    'margin_preset' => 'recommended', // or 'minimum'
    'paper' => 'white',               // cream, etc. -
  ],
  metadata only
],
```

Bleed

Bleed expands the page and margins:

- Page width / height each increase by $2 \times \text{bleed_mm}$
- Every margin increases by bleed_mm

Default `bleed_mm` is 3.

Margin presets (mm, before bleed)

| Preset | Left | Right | Top | Bottom |
|-------------|------|-------|------|--------|
| recommended | 12.7 | 9.525 | 12.7 | 12.7 |

| Preset | Left | Right | Top | Bottom |
|----------------|-------|-------|-------|--------|
| minimum | 9.525 | 6.35 | 9.525 | 9.525 |

Unknown preset names fall back to **recommended**.

Paper

print.paper is written into the KDP metadata sidecar. It does not change PDF colour or stock — choose white vs cream in the KDP dashboard to match.

Trim size

Uses the same `document.size` / `document.format` as your regular PDFs. `papyrus doctor` warns when trim falls outside KDP bounds (width 101.6–215.9 mm, height 152.4–279.4 mm). List presets with `papyrus sizes`.

Cover

The print PDF **omits** the cover page. Export wraparound / ebook cover files with **kdp:cover**.

kdp:cover

Copy configured cover images from **assets/** into **export/** with KDP-oriented filenames. No enable flag — safe to run anytime.

```
papyrus kdp:cover  
papyrus kdp:cover -d /path/to/book -e /path/to/out
```


Options

| Option | Short | Default | Meaning |
|-----------------------|-----------------|----------------------|------------------|
| <code>--dir</code> | <code>-d</code> | current directory | Book root |
| <code>--export</code> | <code>-e</code> | <code>export/</code> | Output directory |

Sources and outputs

| Source | Export name |
|---------------------------------------|---|
| kdp.ebook.cover
→
assets/<file> | <slug>-kdp-ebook-cover.<ext> |
| cover[<theme>]
or cover.image | <slug>-kdp-print-cover-<theme>.<ext>
(one per theme) |

Missing source files are skipped (no error). If nothing is copied, the command still succeeds with a comment.

Example

```
'cover' => [  
  'image' => 'cover.png',  
  'light' => 'cover-light.png',  
],  
'kdp' => [  
  'ebook' => [  
    'cover' => 'cover-ebook.jpg',  
  ],  
],
```

With themes **light** and **dark**, a typical export folder might contain:

```
export/my-book-kdp-ebook-cover.jpg  
export/my-book-kdp-print-cover-light.png  
export/my-book-kdp-print-cover-dark.png # if dark cover  
resolves
```

Papyrus copies files as-is — it does not generate a full wraparound print cover. Use Amazon's cover calculator for spine width and bleed on the print jacket.

kdp:metadata

Emit a JSON sidecar you can keep next to KDP uploads or feed into your own publishing scripts. Always writes — no enable flag.

```
papyrus kdp:metadata  
papyrus kdp:metadata -d /path/to/book -e /path/to/out
```

Options

| Option | Short | Default | Meaning |
|-----------------------|-----------------|----------------------|------------------|
| <code>--dir</code> | <code>-d</code> | current directory | Book root |
| <code>--export</code> | <code>-e</code> | <code>export/</code> | Output directory |

Output

```
export/<slug>-kdp-metadata.json
```

Config

```
'language' => 'en',

'kdp' => [
  'print' => [
    'paper' => 'white',
    'bleed_mm' => 3,
    'margin_preset' => 'recommended',
  ],
  'metadata' => [
    'description' => 'Bookstore blurb...',
    'keywords' => ['papyrus', 'markdown', 'ebook'],
    'language' => 'en', // optional; else project language
  ],
],
```

JSON shape

```
{
  "title": "...",
  "subtitle": "...",
  "author": "...",
  "language": "en",
  "description": "...",
  "keywords": [...],
  "print": {
    "paper": "white",
    "bleed_mm": 3,
    "margin_preset": "recommended"
  }
}
```

| Field | Source |
|--|--|
| <code>title</code> / <code>subtitle</code> / <code>author</code> | Project identity |
| <code>language</code> | <code>kdp.metadata.language</code> → <code>language</code> → <code>en</code> |
| <code>description</code> | <code>kdp.metadata.description</code> (may be empty) |
| <code>keywords</code> | <code>kdp.metadata.keywords</code> (string list) |

| Field | Source |
|----------------------|---|
| <code>print.*</code> | Current <code>kdp.print</code> settings |

This file is a helper for your workflow — it is not uploaded automatically to Amazon.

Migration and CI

Migrating from ibis-next

```
papyrus migrate-ibis
papyrus migrate-ibis -d /path/to/book
papyrus migrate-ibis --force
```

| Option | Short | Meaning |
|----------------------|-----------------|--|
| <code>--dir</code> | <code>-d</code> | Book root |
| <code>--force</code> | <code>-f</code> | Overwrite an existing <code>papyrus.php</code> |

This writes `papyrus.php` from `ibis.php` and updates theme TOC markers to `<!-- PAPYRUS:TOC -->`. After migration:

1. Remove `hi-folks/ibis-next` and any Composer patches for `ibis/mPDF`.
2. Point scripts at `papyrus` (`build`, `build:pdf`, `build:site`, ...).
3. Run `papyrus doctor` and a full `papyrus build`.

Continuous integration

Copy the stub workflow from the Papyrus package:

```
stubs/github/workflows/book-build.yml
```

A typical book job installs PHP with `dom`, `gd`, `mbstring`, `zip`, `zlib`, runs `composer install`, `papyrus doctor`, optional `papyrus lint`, then `composer build`.

To publish a `build:site` output on GitHub Pages, add a job that runs `papyrus build:site` and deploys the site directory to a `gh-pages` branch (see [pages.yml](#) in this repository for a working example, including Chromium for Mermaid).

Programmatic use

```
use Milton\Papyrus\Config\Project;

$project = Project::load($bookDir);
$book =
$project->bookConverter()->convertDirectory($project->contentDir)
;
```

Checklist before release

1. `papyrus doctor` is clean
2. `papyrus build` succeeds for every theme
3. `papyrus build:site` looks right on phone and desktop (try light and dark)
4. Sample ranges still make sense after reflows (`build:sample`)
5. KDP EPUB opens in Kindle Previewer; print PDF matches trim/bleed
6. Pin a Papyrus version in the book's `composer.json`

Report problems at github.com/milon/papyrus/issues.

Command reference

Build

| Command | Purpose | Notable options |
|---------------------------|--|--|
| <code>build</code> | PDF themes, EPUB, HTML, enabled KDP | <code>-d</code> , <code>-e</code> |
| <code>build:pdf</code> | PDF themes | <code>--theme</code> , <code>--parallel</code> |
| <code>build:epub</code> | EPUB3 | <code>-d</code> , <code>-e</code> |
| <code>build:html</code> | Single-file HTML | <code>-d</code> , <code>-e</code> |
| <code>build:site</code> | Multi-page HTML site | <code>-d</code> , <code>-e</code> |
| <code>build:sample</code> | Sample PDF from <code>sample.ranges</code> | <code>--theme</code> |

KDP

| Command | Purpose | Notable options |
|---------------------------|---|--|
| <code>kdp</code> | All enabled KDP outputs | <code>-d</code> , <code>-e</code> |
| <code>kdp:ebook</code> | Kindle EPUB | requires <code>kdp.ebook.enabled</code> |
| <code>kdp:print</code> | Print interior PDF | <code>--theme</code> ; requires <code>kdp.print.enabled</code> |
| <code>kdp:cover</code> | Copy cover assets to <code>export/</code> | no enable flag |
| <code>kdp:metadata</code> | Metadata JSON sidecar | no enable flag |

Project tooling

| Command | Purpose | Notable options |
|---------------------------|--|--|
| <code>init</code> | Scaffold <code>papyrus.php</code> , <code>content/</code> , <code>assets/</code> | <code>--force</code> |
| <code>doctor</code> | Validate config, paths, Mermaid, KDP trim | <code>-d</code> |
| <code>sizes</code> | List page-size presets (+ KDP in-bounds) | (no <code>-d</code> / <code>-e</code>) |
| <code>lint</code> | Lint PHP fences in <code>content/</code> | <code>--fix</code> , <code>--max-width=66</code> |
| <code>watch</code> | Rebuild via <code>build</code> on file changes | <code>--interval</code> |
| <code>migrate-ibis</code> | <code>ibis.php</code> → <code>papyrus.php</code> | <code>--force</code> |

Shared book options

Most book commands accept:

| Option | Short | Default | Meaning |
|-----------------------|-----------------|----------------------------------|--|
| <code>--dir</code> | <code>-d</code> | current directory | Book root (contains <code>papyrus.php</code>) |
| <code>--export</code> | <code>-e</code> | <code><book>/export</code> | Override export directory |

Also useful globally: `-v` / `--verbose` (include third-party vendor notices), `-q` / `--quiet`, `--no-ansi`.

Output filenames

| Command | Path under export |
|---------------------------|---|
| <code>build:pdf</code> | <code><slug>-<theme>.pdf</code> |
| <code>build:html</code> | <code><slug>.html</code> |
| <code>build:epub</code> | <code><slug>.epub</code> |
| <code>build:site</code> | <code><slug>-site/</code> |
| <code>build:sample</code> | <code>sample-<slug>-<theme>.pdf</code> |
| <code>kdp:ebook</code> | <code><slug>-kdp.epub</code> |
| <code>kdp:print</code> | <code><slug>-kdp-print.pdf</code> |
| <code>kdp:cover</code> | <code><slug>-kdp-ebook-cover.*, <slug>-kdp-print-cover-<theme>.*</code> |
| <code>kdp:metadata</code> | <code><slug>-kdp-metadata.json</code> |

Packagist: [milon/papyrus](https://packagist.org/packages/milon/papyrus). Source: github.com/milon/papyrus. Releases: github.com/milon/papyrus/releases.

Downloads

Preview this handbook as a PDF without building locally. The files live in the Papyrus repository under [docs/](#) and update when the handbook is rebuilt.

PDF

| Theme | Download |
|-------|---|
| Light | <u>the-papyrus-handbook-light.pdf</u> |
| Dark | <u>the-papyrus-handbook-dark.pdf</u> |

Open the link to download from GitHub, or use **View raw** / **Download** on the file page:

- [Light PDF on GitHub](#)
- [Dark PDF on GitHub](#)

Other previews

| Format | Link |
|------------------|--|
| This site | <u>milon.im/papyrus</u> |
| Single-file HTML | <u>the-papyrus-handbook.html</u> |

Rebuild everything from the Papyrus repo with:

```
composer build:handbook
```